

Operator & external-agent entry surfaces — all pass the same binding model

Web console

TUI workbench

MCP server · 25 tools

A2A endpoint

Plugin host (hot-reload)

Agent half

Harness registry — 14 engines
three-stage availability probe

Teams & channels
lead supervision · task tree · watchdog

Memory stack
FTS5 + vector · RRF + MMR · compaction

Governed bridge — the only path to actuation

proposals

Agent tool surface — 48 tools
semantic cards ground physical units

Node bindings
manual / auto / unbound gate the path

Governed write path (Alg. 1)
approval → guardrail → interlock →
readback-verified write → journal

Backstop (30 s cadence, breach ≥ 3):
bounded rollback K=2 → human

journal · state

verified write
↑ readback

Industrial half

Protocol drivers
Modbus TCP/RTU · OPC UA · MQTT · HTTP/REST

Edge runtimes — 3 decoupled intervals
per-node schedule · in-flight exclusion

Gateway sweep
rotating cursor · starvation-free quota 64

Alarm chain
hysteresis · debounce · recipe-window overlay

Line / recipe management
versioning · known-good · active runs

film-line
scenario

cast-film physics
twin + ODE

WS push

scene-edit writeback

Audit plane

signed journal · ops log · attribution

Storage — one data domain

TimescaleDB + MinIO · SQLite (WAL, FTS5)

Device-twin registry

modelRef · boundAgentId · telemetry / desired / state

3D scene (three.js)

live render · twin observes

Data, audit & digital-twin plane — one data domain